ProtSCAPE: Mapping the landscape of protein conformations in molecular dynamics

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Abstract

Proteins are inherently dynamic molecules whose biological functions depend on transitions among multiple conformational states. Although recent machine learning methods have achieved remarkable success in predicting static protein structures, accurately modeling conformational ensembles and time-dependent dynamics remains a major challenge. Molecular dynamics (MD) simulations provide atomistic trajectories of protein motion, but generating sufficiently long and densely sampled trajectories is computationally expensive. In this work, we introduce and evaluate ProtSCAPE, a geometric deep learning framework that learns latent representations of protein conformations and generates new structures through diffusion-based sampling in its latent space, as a method for reconstructing dense ensembles from sparsely sampled MD data.

We study an upsampling setting in which ProtSCAPE is trained on only every tenth frame of reference MD trajectories and tasked with generating full ensembles for five benchmark protein systems. Performance is assessed using both structural quality metrics, including MolProbity score, clash score, and Ramachandran outliers, and ensemble-level agreement metrics, including RMSF correlation, pairwise RMSD, Root Mean Wasserstein Distance (RMWD), and PCA-space Wasserstein distances. Across all systems, ProtSCAPE consistently produces stereochemically plausible structures while closely matching the equilibrium distributions and collective motions of the original MD simulations. In many cases, it outperforms competing generative baselines including AlphaFLOW and MDGen.

These results demonstrate that ProtSCAPE can efficiently recover realistic protein conformational landscapes from limited trajectory data, highlighting the promise of latent generative models for scalable analysis and acceleration of biomolecular simulation.

1 Introduction

Proteins are inherently dynamic molecules whose biological functions are intimately tied to motions occurring across a broad range of spatial and temporal scales. While recent breakthroughs in machine learning (ML) have enabled remarkably accurate prediction of static folded protein structures from sequence alone, these approaches largely capture equilibrium conformations and provide limited insight into conformational heterogeneity and time-dependent structural rearrangements [1, 2].

Molecular dynamics (MD) is a computational technique in which the equations of motion are integrated for a system of atoms and molecules, generating a time-resolved trajectory of coordinates. By explicitly evolving the positions and momenta with a timestep on the order of femtoseconds, MD can provide temporal insight into biophysical structural fluctuations and chemical reactions that are otherwise inaccessible via traditional experimental methods. Such simulations offer a physics-based framework for probing protein motion at atomic resolution, but the resulting trajectories span high-dimensional conformational spaces [3]. Extracting interpretable, global representations of protein dynamics from such data remains a challenge, as many existing analysis techniques rely on low-dimensional projections or chemist-designed collective variables that may require significant *a prior* knowledge of a specific biological system [4].

In addition, the high time resolution that makes MD a powerful tool also imposes a limitation on the total simulation time that can be reached with current computational resources. Many biophysical phenomena such as protein folding, ligand binding, conformational transitions of nucleic acids, and larger scale domain motions occur on micro to millisecond timescales, orders of magnitude longer than the nanosecond to microsecond windows routinely attainable by MD without acceleration [3].

To address these challenges, the Krishnaswamy lab is currently developing the Protein Transformer with Scattering Attention and Positional Embedding (ProtSCAPE) model, a deep learning framework that combines learnable geometric scattering transforms with transformer-based attention mechanisms to model protein dynamics from MD trajectory data [5]. ProtSCAPE has already demonstrated strong performance in learning structured, interpretable latent representations of protein conformational dynamics [5]. This continuous lower-dimensional latent space enables interpolation, generalization across trajectories, and mechanistically plausible reconstructions from MD data.

However, while these results highlight the model's representational power, its evaluation has focused primarily on geometric and structural accuracy rather than on consistency with the corresponding MD ensembles and transition pathways. Incorporating more rigorous biophysical validation such as free-energy landscape agreement and pathway-level rare-event analysis would strengthen confidence in ProtSCAPE's physical realism and motivate the proposed work to more deeply assess and extend its capabilities.

2 Background

Fundamentally, biophysical systems such as proteins and other macromolecules are dynamic rather than static objects, continuously fluctuating among many conformations. The behavior of these systems at equilibrium is naturally described by an underlying free-energy surface (FES), a high-dimensional landscape that assigns an effective free energy to each possible molecular configuration, providing a coarse-grained description of stability.

Under equilibrium conditions, the probability of observing any given conformation is proportional to the Boltzmann weight, i.e. the negative exponential of its free energy. Consequently, low-free-energy regions are sampled more frequently than high-free-energy regions. Typical FES of proteins are characterized by deep basins corresponding to metastable conformational states, connected by saddle regions or separated by energy barriers whose crossings occur as rare transition events [6, 7]. These basins often correspond to biologically meaningful structures such as folded states, intermediate states, bound complexes, or alternative functional conformations.

This dynamic perspective highlights an important limitation of standard single-structure protein models. Experimental structures such as X-ray crystal conformations, as well as predictive methods such as AlphaFold2 and AlphaFold3 [8, 1], typically return one dominant or representative structure. While highly valuable, such approaches capture only a small subset of the accessible conformational landscape, often corresponding to one low-energy basin within a much larger FES. As a result, they may miss alternative states and transition pathways that are central to biological function.

2.1 Related Work

A wide range of computational approaches have been developed to study protein structure and interactions, enabling advances in sequence-based structure prediction, stability assessment, and the analysis of biomolecular binding. By comparison, relatively fewer machine learning methods have focused on leveraging MD simulations to learn and characterize protein conformational dynamics directly.

Recent work has increasingly explored generative approaches to modeling protein conformational variability and dynamics. AlphaFLOW is a generative protein-structure modeling framework that extends AlphaFold to predict not just one static structure, but an ensemble of plausible conformations for a protein [9]. The method combines AlphaFold's single-structure accuracy with flow matching, a generative modeling technique that learns how to transform a noise distribution into the desired ensemble distribution. During sampling, the model iteratively refines an initial polymer-like random structure

through multiple denoising steps, using AlphaFold's structure reasoning network as the learned flow field. After denoising, the distribution of generated protein structures emulates the that of data distribution, allowing AlphaFLOW to generate diverse but physically coherent structures conditioned on sequence.

Similarly, MDGen learns conditional generative models of full molecular dynamics trajectories, allowing direct generation, interpolation, and upsampling of protein motions as a surrogate for conventional MD simulation [10]. Architecturally, MDGen represents trajectories as arrays of tokens that encode residue roto-translation offsets relative to key frames plus torsion angles across time. These tokens are modeled with a flow-based transformer backbone using stochastic interpolants, allowing the system to generate trajectories from Gaussian noise while conditioning on known frames, residues, or sequence information. The model uses alternating attention across residues and time, enabling it to capture both spatial molecular interactions and temporal dynamics efficiently. While MDGen was mostly designed for and tested on small peptides, it is capable of modeling full protein systems as well [10]. The authors emphasize MDGen's ability to perform prediction in a variety of regimes, including data upsampling, in which sparsely sampled MD frames can be interpolated to infill a full ensemble of structures [10]. This work will focus on this usecase in its comparisons to the ProtSCAPE architechture.

2.2 ProtSCAPE Architecture

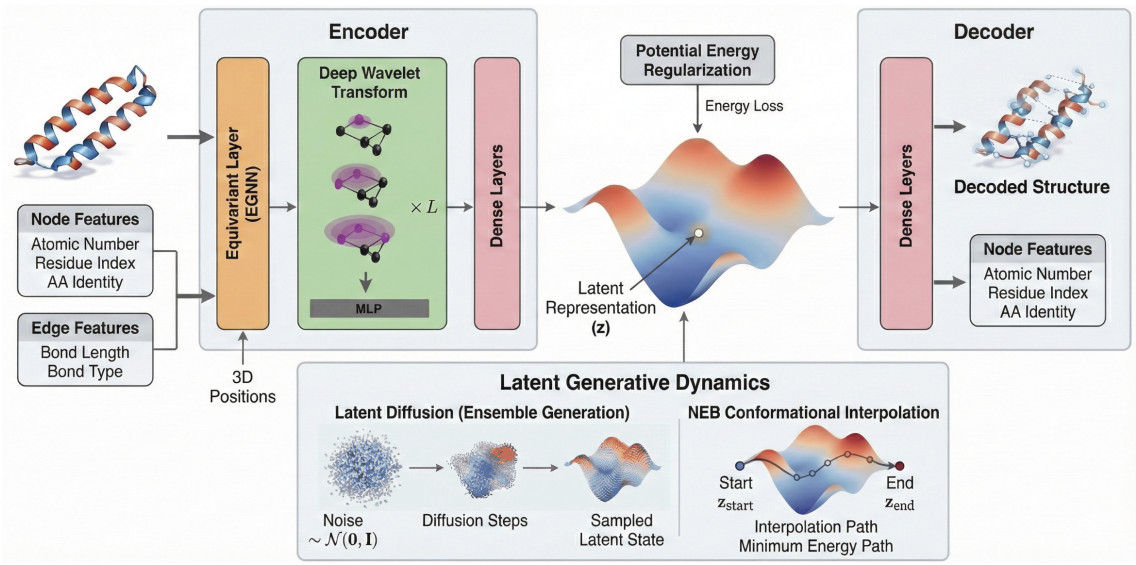

Figure 2.1: Schematic of the ProtSCAPE model architecture.

ProtSCAPE by contrast adopts a representation learning framework to model protein conformational dynamics by learning a low-dimensional latent space that can be decoded into protein structures and is explicitly organized along the temporal progression of molecular dynamics trajectories, enabling smooth interpolation between conformational

states [5]. It is an encoder-decoder framework that learns a compact map of protein conformations from MD trajectories (Figure 2.1). Each protein structure is represented as a graph, with atoms or residues as nodes and bonds or spatial relationships as edges, then passed through an equivariant geometric network that respects rotational and translational symmetry. Importantly, it incorporates deep graph wavelet transforms, which replaces ordinary local message passing with multiscale diffusion filters. By applying powers of a graph diffusion operator and taking differences between scales, ProtSCAPE separates local structural signals from long-range collective motions, much like wavelets separate frequencies in classical signal processing. This is especially useful for proteins, where biologically important behavior often depends on coordinated changes across multiple scales, from side-chain fluctuations to loop motions and large domain hinge rearrangements [5].

The resulting latent space is an organized approximation of the protein's conformational landscape. A key part of this latent space is that ProtSCAPE regularizes it utilizing molecular mechanics energies so that nearby latent points correspond to energetically similar states. This makes the latent manifold ideal for diffusion modeling, because sampling in a low-dimensional smooth space is far easier than generating directly in thousands of atomic coordinates with geometric constraints. Thus ProtSCAPE uses a denoising diffusion probabilistic model in its learned latent space rather than directly on atomic coordinates. During training, Gaussian noise is gradually added to latent embeddings of real protein conformations, and a neural network learns to predict and remove that noise at each timestep. During generation, the process starts from random Gaussian noise and iteratively denoises it into a valid latent point that lies on the learned conformational manifold, which is then decoded into a 3D protein structure. Because diffusion occurs in the compact latent space, sampling is more stable and efficient than operating in the full high-dimensional coordinate space.

The same latent space also supports transition-path generation, since smooth trajectories between latent points can be decoded into realistic structural intermediates. In effect, ProtSCAPE turns raw MD trajectories into a navigable energy-aware latent space where both equilibrium sampling and conformational transitions become more tractable than in the ambient space of 3-dimensional atomic coordinates.

2.3 Current Goals

Prior work has established ProtSCAPE's capabilities in generating structurally correct protein conformations when compared to other models such as the aforementioned AlphaFLOW and MDGen [5]. This means that sampled proteins are not only diverse but also stereochemically plausible. Evaluating structures using standard validation metrics such as Ramachandran outliers, rotamer outliers, clash scores, and aggregate MolProbity measures. These metrics test whether backbone dihedral angles fall in allowed regions, side chains adopt realistic rotamers, and atoms avoid steric overlaps. Strong performance on these criteria suggests that decoded samples preserve realistic protein geometry rather than matching global shapes.

However, so far there has been a lack of investigation into whether ProtSCAPE's gen-

erated ensembles match the MD-generated ensembles in a distributional sense. More specifically, we want to show that when treating MD structures as samples from a distribution from a FEL, the samples generated by ProtSCAPE come from as similar of a distribution as possible. This work focuses on evaluating ProtSCAPE's ensemble generation capabilities through distributional similarity with MD baselines.

3 Methodology

To quantify agreement between ProtSCAPE predicted and reference MD ensembles, we adopt distributional metrics that assess both atom-wise positional statistics and global collective motions. Because the space of atomic coordinates for all protein atoms is too high-dimensional to obtain sufficient sampling to meaningfully compare distributions, typical analysis turns to lower dimensional projections such as principal-component analysis (PCA) of key features such as Cα positions and dihedral angles. Other metrics of diversity typically highlight overall structural similarity through root mean methods.

3.1 Pairwise RMSD

To quantify the overall structural diversity of an ensemble and assess whether predicted models reproduce the global amplitude of conformational fluctuations observed in MD, we compute the average pairwise root-mean-square deviation (RMSD) between conformations. For an ensemble of predicted structures $X = \{x^{(k)}\}_{k=1}^{K}$, where each conformation consists of the Cartesian coordinates of N atoms (typically only using Cα atoms after alignment), the pairwise RMSD between two static conformations $x^{(k)}$ and $x^{(\ell)}$ is defined as

$$\mathrm{RMSD}(x^{(k)}, x^{(\ell)}) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left\|x_i^{(k)} - x_i^{(\ell)}\right\|_2^2}$$

The ensemble-level pairwise RMSD is then computed as the average over all distinct pairs of structures:

$$\mathrm{pRMSD}(X) = \frac{2}{K(K-1)}\sum_{1\le k<\ell\le K}\mathrm{RMSD}(x^{(k)}, x^{(\ell)}).$$

An analogous quantity $\mathrm{pRMSD}(Y)$ is defined for the reference ensemble $Y = \{y^{(k)}\}_{k=1}^{L}$. To evaluate whether a predicted ensemble reproduces the relative diversity and flexibility for a given system, we can calculate the relative or percent difference between the reference $\mathrm{pRMSD}(Y)$ and predicted $\mathrm{pRMSD}(X)$. This percent difference in pairwise RMSD between predicted and ground truth provides a global measure of ensemble spread by averaging structural dissimilarities across all conformations. It is therefore sensitive to the overall magnitude of conformational variability but does not encode spatial localization of flexibility.

3.2 RMWD

Following prior ensemble-evaluation protocols [9], we also compute the Root Mean Wasserstein Distance (RMWD) across atomic positional distributions and the 2-Wasserstein (W_2) distance between projected ensemble distributions. We employ the W_2 distance rather than standard Kullback-Leibler (KL) divergence to compare structural ensembles because Wasserstein distance provides a more geometrically meaningful and numerically stable measure of distributional discrepancy in high-dimensional spaces typically encountered in protein ensembles. KL divergence also requires overlapping support between distributions and becomes infinite when predicted ensembles fail to populate low-probability regions sampled by MD, a common occurrence in finite structural datasets. In contrast, W_2 remains well-defined for non-overlapping empirical distributions and directly incorporates the underlying Euclidean geometry of configuration space, measuring the minimal transport cost required to transform one ensemble into another.

Let $X = \{x^{(k)}\}_{k=1}^{K}$ denote the predicted ensemble and $Y = \{y^{(k)}\}_{k=1}^{L}$ the reference MD ensemble, each consisting of N atoms. For atom i, we model its positional distribution across conformations as a multivariate Gaussian $\mathcal{N}(\mu_i^X, \Sigma_i^X)$, $\mathcal{N}(\mu_i^Y, \Sigma_i^Y)$, where μ_i and Σ_i are the empirical mean and covariance of the 3D coordinates of atom i across the ensemble. In practice, each residue is represented by its Cα position, modeling each as an independent distribution. Then the squared 2-Wasserstein distance between two multivariate Gaussian distributions admits the closed form:

$$W_2^2(\mathcal{N}(\mu_1, \Sigma_1), \mathcal{N}(\mu_2, \Sigma_2)) = \|\mu_1 - \mu_2\|_2^2 + \mathrm{Tr}\left(\Sigma_1 + \Sigma_2 - 2\left(\Sigma_2^{1/2}\Sigma_1\Sigma_2^{1/2}\right)^{1/2}\right).$$

We can then define the (squared) Root Mean Wasserstein Distance for our prediction enssemble X and reference ensemble Y as

$$\begin{aligned}\mathrm{RMWD}^2(X, Y) &= \frac{1}{N}\sum_{i=1}^{N} W_2^2\big(\mathcal{N}(\mu_i^X, \Sigma_i^X), \mathcal{N}(\mu_i^Y, \Sigma_i^Y)\big) \\ &= \underbrace{\frac{1}{N}\sum_{i=1}^{N}\|\mu_{1,i} - \mu_{2,i}\|^2}_{\text{(translation contribution)}^2} + \underbrace{\frac{1}{N}\sum_{i=1}^{N}\mathrm{Tr}\left(\Sigma_{1,i} + \Sigma_{2,i} - 2\left(\Sigma_{1,i}^{1/2}\Sigma_{2,i}\Sigma_{1,i}^{1/2}\right)^{1/2}\right)}_{\text{(variance contribution)}^2}\end{aligned}$$

In the limit where each ensemble contains a single structure, RMWD reduces to the standard RMSD which is typically used to measure differences between static structures. As noted in the equation above, the metric naturally decomposes into contributions from mean structural shifts ($\|\mu_i^X - \mu_i^Y\|^2$) and covariance discrepancies (flexibility differences encoded in Σ_i), thereby separating errors in equilibrium structure from errors in atomic fluctuations.

3.3 PCA W_2 Distance

To additionally assess ensemble agreement at the level of collective motions, we compare the distributions of conformations projected onto a low-dimensional principal component subspace. Principal component analysis PCA is first performed on the reference MD ensemble (using Cα coordinates or dihedral angles ϕ, ψ for each amino acid), yielding orthonormal components $\{v_j\}_{j=1}^{d}$. Both predicted and reference ensembles are projected onto the first two principal components v_1, v_2:

$$z^{(k)} = \left(\langle x^{(k)}, v_1\rangle, \langle x^{(k)}, v_2\rangle\right).$$

where $\langle \cdot, \cdot \rangle$ represents the standard inner dot product. Let $\mathcal{P}_X$ and $\mathcal{P}_Y$ denote the empirical distributions of projected coordinates for the predicted and MD ensembles, respectively. The ensemble discrepancy in collective space is quantified via the 2-Wasserstein distance $W_2(\mathcal{P}_X, \mathcal{P}_Y)$. In practice, the projected distributions are either approximated as Gaussians (using empirical mean and covariance in PCA space) or computed using discrete optimal transport between empirical samples at a higher computational expense.

3.4 RMSF Correlation

To further evaluate whether predicted ensembles reproduce the spatial pattern of residue-level flexibility observed in MD, we compute the correlation between root-mean-square fluctuations (RMSF) profiles. For each residue (or Cα atom) i, the RMSF is defined as the square root of the variance of its positional distribution across the ensemble:

$$\mathrm{RMSF}_i^X = \sqrt{\frac{1}{K}\sum_{k=1}^{K}\left\|x_i^{(k)} - \mu_i^X\right\|_2^2}$$

where μ_i^X and μ_i^Y denote the mean positions of atom i in the predicted and reference ensembles, respectively. The agreement between flexibility profiles between the reference ensemble Y and the predicted ensemble X is then quantified using the Pearson correlation coefficient

$$\rho_{\mathrm{RMSF}}(X,Y) = \frac{\sum_{i=1}^{N}\left(\mathrm{RMSF}_i^X - \overline{\mathrm{RMSF}^X}\right)\left(\mathrm{RMSF}_i^Y - \overline{\mathrm{RMSF}^Y}\right)}{\sqrt{\sum_{i=1}^{N}\left(\mathrm{RMSF}_i^X - \overline{\mathrm{RMSF}^X}\right)^2}\sqrt{\sum_{i=1}^{N}\left(\mathrm{RMSF}_i^Y - \overline{\mathrm{RMSF}^Y}\right)^2}}$$

where $\overline{\mathrm{RMSF}^X}$ and $\overline{\mathrm{RMSF}^Y}$ denote the mean RMSF values across all residues. Unlike RMWD, which captures both positional and covariance discrepancies in an absolute sense, RMSF correlation provides a scale-invariant measure of whether the predicted ensemble correctly identifies regions of high and low flexibility along the sequence. As such, it is particularly sensitive to the relative distribution of fluctuations, complementing transport-based metrics by isolating agreement in fluctuation patterns independent of global magnitude differences.

3.5 Training Procedures

In order to evaluate ProtSCAPE's ensemble generation capabilities, we emphasize the usecase of upsapmling as a case study for several protein systems. In this regime, the model is exposed to a subsampled version of the true ground truth, typically $0.1 - 0.01$ of the full MD trajectory, and then predicts a the full ensemble of structures. This replicates the common problem in MD where running a full trajectory or even storing the full data associated with a longer trajectory can cause significant computational overhead. Being able to take a subsample of the true MD trajectory and interpolate into a richer ensemble that still reflects equilibrium dynamics all at low cost is of great analytical value when less compute is available.

In this case study, we trained the ProtSCAPE model on trajectories from the ATLAS MD database for a set of representative protein systems: `6h86_A`, `6p5h_A`, `6jv8_A`, `7lp1_A`, and `7jfl_C`, where each identifier denotes a specific Protein Data Bank (PDB) structure code and chain. For each system, ATLAS provides three replicates of trajectory data, each corresponding of 100ns of chemical time, corresponding to approximately 10,000 frames after preprocessing. To simulate the desired upsampling conditions, the model was trained on subset of the first replicate consisting of every 10th frame or 1,000 frames total. We trained each instance of ProtSCAPE for 1500 epochs in total or around 2 hours of wall-clock time.

Following training, ProtSCAPE was used to generate ensembles of full 10,000 structures ensembles via diffusion-based sampling in the learned latent space. he resulting ensembles were evaluated against the reference MD trajectories using the ensemble consistency metrics described above in addition to the standard structural correctness metrics employed previously. Comparative benchmarking against alternative generative ensemble models, including AlphaFlow and MDGen was also run, using the same upsampling conditions mentioned previously.

3.6 Baseline Comparisons

To serve as baseline comparisons to the state-of-the art in ensemble generation, we compared the ProtSCAPE model to the previously discussed AlphaFLOW and MDGen [9, 10]. Though these models were designed for a more transferable usecase, they were trained in the same way as ProtSCAPE for the upsampling paradigm, with exposure to a training set of 1,000 out of the total 10,000 frames for the first replicate in each ATLAS system.

4 Results

4.1 Structural Correctness

In line with previous work, we tested for baseline structural correctness of the generated ensembles. Ramachandran outliers measure how often backbone ϕ and ψ dihedral angles fall in sterically disallowed regions, while clash score quantifies steric overlaps between atoms that are too close together, indicating physically implausible packing. This is further combined into an overall MolProbity score, a widely used summary metric of stereochemical quality, where lower scores indicate more realistic protein structures [11].

The results for these calculations are listed in table 4.1. We found that ProtSCAPE consistently produces high-quality structural predictions, with the lowest average MolProbity score in all five systems tested and the lowest average clash score in four of the five. While the AlphaFLOW model generally outperformed ProtSCAPE in terms of the percentage of Ramachandran outliers, ProtSCAPE still retains a low number of outliers, with no system exceeding 1.3% on average.

This data suggests that ProtSCAPE achieves a balance between local stereochemical accuracy and global structural realism. In particular, the consistently low MolProbity scores indicate that structures generated through latent-space diffusion remain well packed and geometrically plausible after decoding back to Cartesian coordinates. This is notable because generative models that emphasize ensemble diversity often risk producing distorted backbone geometries or steric clashes when sampling away from the training manifold. ProtSCAPE appears to avoid this failure, implying that its learned latent representation preserves physically meaningful constraints during generation.

By contrast, MDGen exhibited substantially worse values across all structural metrics, including very high clash scores and elevated Ramachandran outlier percentages. This suggests that although trajectory-level generation may capture certain dynamic features, direct generative modeling of coordinates can be more prone to local geometric artifacts under the present upsampling setting. AlphaFLOW remained competitive, especially in backbone dihedral quality, but ProtSCAPE's superior MolProbity performance across all systems indicates stronger overall stereochemical consistency.

We do note that because ProtSCAPE does not predict the side chain structures that MDGen and AlphaFLOW do, this gives it an advantage in generating structures that do not produce steric clashes and also are more likeley to have a lower MolProbity score. Ramachandran outliers remain a standard baseline for all three models however.

Overall, these findings demonstrate that ProtSCAPE generally does not trade physical plausibility for conformational diversity. It produces ensembles that are structurally valid and suitable for downstream biophysical analyses, providing an essential prerequisite before evaluating higher-order agreement with MD distributions.

Table 4.1: Structural quality metrics across protein datasets. Lower is better for all reported metrics. Best result in each category is bolded. Rotamer outlier statistics are omitted.

Protein	**Method**	Rama. Outliers↓	Clash Score↓	MolProbity Score↓
6H86 (Chain A)	AlphaFlow	0.9864 ± 0.498	**15.1059 ± 5.596**	2.2979 ± 0.181
	MDGen	29.9647 ± 9.691	159.3615 ± 35.461	3.9921 ± 0.277
	ProtSCAPE	**0.5639 ± 0.945**	18.2541 ± 18.416	**1.6399 ± 0.488**
6P5H (Chain A)	AlphaFlow	**0.4074 ± 0.666**	16.7337 ± 5.121	1.9246 ± 0.227
	MDGen	10.3405 ± 3.443	164.8066 ± 24.569	3.9622 ± 0.192
	ProtSCAPE	1.2644 ± 1.122	**16.2129 ± 12.608**	**1.8516 ± 0.467**
6JV8 (Chain A)	AlphaFlow	**1.1344 ± 1.157**	19.0241 ± 6.913	1.9835 ± 0.248
	MDGen	17.8680 ± 5.696	161.0425 ± 26.028	3.9382 ± 0.230
	ProtSCAPE	1.2751 ± 1.548	**16.4361 ± 15.736**	**1.6672 ± 0.599**
7LP1 (Chain A)	AlphaFlow	**0.9238 ± 1.432**	30.2362 ± 11.954	2.7960 ± 0.317
	MDGen	17.7048 ± 8.142	214.6923 ± 43.647	4.3451 ± 0.236
	ProtSCAPE	1.3025 ± 2.148	**23.9564 ± 23.546**	**1.9311 ± 0.691**
7JFL (Chain C)	AlphaFlow	**0.1878 ± 0.910**	20.4902 ± 9.839	1.9722 ± 0.321
	MDGen	50.1211 ± 10.584	169.6357 ± 30.496	4.2805 ± 0.244
	ProtSCAPE	0.6950 ± 1.286	**11.3824 ± 16.178**	**1.3773 ± 0.640**

4.2 Ensemble Consistency

While structural correctness ensures that individual generated conformations are chemically plausible, as mentioned in the methodology section, it does not establish whether a model reproduces the distribution of states sampled by MD. For many biological applications, the central objective is not just to generate valid static structures, but to recover the equilibrium ensemble, including the relative populations of metastable states and overall amplitudes of motion. We therefore evaluated ensemble fidelity using the metrics in Tables 4.2 and 4.3.

4.2.1 Dynamic Agreement Metrics

We first examined RMSF correlation and pairwise RMSD percent difference, which quantify residue-level flexibility patterns and overall ensemble spread, respectively (Table 4.2). ProtSCAPE achieved the highest RMSF correlation in all five systems, with values ranging from 0.9988 to 0.9998. These near-unity correlations indicate that ProtSCAPE accurately reproduces which regions of each protein are rigid versus flexible, closely matching the fluctuation profiles observed in the reference MD trajectories.

This result is particularly significant because RMSF reflects collective equilibrium be-

havior accumulated over thousands of frames rather than instantaneous geometric similarity. Recovering these profiles suggests that ProtSCAPE captures physically meaningful dynamical heterogeneity rather than simply memorizing static conformations.

ProtSCAPE also achieved the best pairwise RMSD agreement in three of five systems and remained competitive in the other two. Lower pairwise RMSD error indicates that the overall conformational breadth of the generated ensemble matches the reference trajectory. Generally this means ProtSCAPE correctly estimates how broadly each protein explores configuration space. MDGen slightly outperformed ProtSCAPE on `6JV8` and `7LP1`, but ProtSCAPE remained within a narrow margin while substantially outperforming AlphaFLOW across all systems.

Table 4.2: Dynamic agreement metrics across protein datasets. Higher RMSF correlation is better. Lower pairwise RMSD percent difference is better. Best result in each category is bolded.

Protein	**Method**	RMSF Corr.↑	Pairwise RMSD↓
6H86 (Chain A)	AlphaFlow	0.8965 ± 0.013	0.7875 ± 0.004
	MDGen	0.9964 ± 0.000	0.1141 ± 0.018
	ProtSCAPE	**0.9997 ± 0.000**	**0.0664 ± 0.010**
6P5H (Chain A)	AlphaFlow	0.9980 ± 0.000	0.2234 ± 0.006
	MDGen	0.9979 ± 0.000	0.1082 ± 0.016
	ProtSCAPE	**0.9998 ± 0.000**	**0.0403 ± 0.015**
6JV8 (Chain A)	AlphaFlow	0.9971 ± 0.000	0.3845 ± 0.007
	MDGen	0.9986 ± 0.000	**0.0332 ± 0.011**
	ProtSCAPE	**0.9994 ± 0.000**	0.0458 ± 0.012
7LP1 (Chain A)	AlphaFlow	0.9548 ± 0.003	0.4331 ± 0.012
	MDGen	0.9932 ± 0.000	**0.0293 ± 0.010**
	ProtSCAPE	**0.9988 ± 0.000**	0.0701 ± 0.008
7JFL (Chain C)	AlphaFlow	0.9888 ± 0.002	0.4351 ± 0.013
	MDGen	0.9985 ± 0.000	0.0448 ± 0.008
	ProtSCAPE	**0.9998 ± 0.000**	**0.0426 ± 0.009**

4.2.2 Distributional Agreement Metrics

To more directly compare equilibrium ensembles, we next considered RMWD and PCA-space Wasserstein distances. These metrics evaluate whether generated samples match the positional means, fluctuations, and collective-state occupancies of the MD reference ensemble.

ProtSCAPE achieved the best RMWD score in all five systems, often by a large margin. For example, on `6H86` ProtSCAPE obtained an RMWD of 0.3441 compared with

1.0066 for MDGen and 4.2487 for AlphaFLOW. Similar trends were observed across every benchmark protein. Because RMWD decomposes into translation and variance terms, this indicates that ProtSCAPE simultaneously captures both average structural positioning and the covariance structure of atomic fluctuations.

The decomposition further clarifies model behavior. ProtSCAPE consistently achieved the lowest translation and variance errors in every system, meaning that it reproduces not only the central tendency of the ensemble but also the magnitude and anisotropy of thermal motion. Matching both components is challenging: a model may predict correct average structures while underestimating flexibility, or generate broad ensembles centered on incorrect means. ProtSCAPE avoids both failure modes.

4.2.3 Collective Motions in PCA Space

Projection into the leading principal-component subspace provides a coarse-grained view of the free-energy landscape, where basins and transition pathways become more interpretable. Across nearly all systems, ProtSCAPE achieved the lowest pooled and reference PCA-W_2 distances, indicating the closest match to MD occupancy distributions in collective-coordinate space.

These results suggest that ProtSCAPE preserves large-scale conformational motions rather than merely matching local residue fluctuations. In other words, the model captures how structures are arranged across dominant slow modes of motion, which are often the biologically relevant degrees of freedom associated with gating, loop rearrangements, or domain motions.

The only partial exception was system 7JFL, where MDGen slightly outperformed ProtSCAPE in PCA-W_2 distance while ProtSCAPE still retained the best RMWD score. This may indicate that MDGen approximated the dominant low-dimensional projection well for this specific system, while ProtSCAPE more accurately matched the full coordinate-space ensemble statistics. Such discrepancies reinforce the value of evaluating multiple complementary metrics.

Overall Interpretation

Taken together, the ensemble-consistency results strongly support the hypothesis that ProtSCAPE is not merely a structure generator, but an effective generative model of molecular dynamics ensembles. Compared with AlphaFLOW and MDGen, ProtSCAPE more faithfully reconstructs equilibrium distributions, reproduces residue flexibility patterns, and captures collective motions across diverse protein systems.

We hypothesize that this advantage arises from ProtSCAPE's architecture. By first learning a smooth, energy-regularized latent manifold of conformations and then performing diffusion in that lower-dimensional space, the model avoids the difficulty of generating directly in high-dimensional Cartesian coordinates. Sampling in latent space likely encourages trajectories that remain on the physically meaningful conformational manifold while still exploring diverse states.

From a practical perspective, these findings suggest that ProtSCAPE is particularly

well suited for trajectory upsampling. Given only a sparse subset of MD frames, the model can reconstruct a dense ensemble that closely resembles the original simulation distribution. This could substantially reduce the storage and computational burden associated with long-timescale MD simulations while preserving biologically relevant ensemble information.

Table 4.3: Distributional ensemble quality metrics across protein datasets. Lower is better for all reported metrics. Best result in each category is bolded.

Protein	**Method**	RMWD↓	Trans.↓	Var.↓	Pool PCA-W_2 ↓	Ref PCA-W_2 ↓
6H86 (Chain A)	AlphaFlow	4.2487 ± 0.017	2.4578 ± 0.005	3.4657 ± 0.021	34.9194 ± 0.195	34.6883 ± 0.195
	MDGen	1.0066 ± 0.008	0.8097 ± 0.008	0.5978 ± 0.014	3.8805 ± 0.209	3.7751 ± 0.207
	ProtSCAPE	**0.3441 ± 0.016**	**0.2070 ± 0.013**	**0.2749 ± 0.010**	**1.4490 ± 0.215**	**1.4576 ± 0.215**
6P5H (Chain A)	AlphaFlow	1.0504 ± 0.061	0.7488 ± 0.077	0.7350 ± 0.010	8.0587 ± 0.839	8.0806 ± 0.830
	MDGen	0.6200 ± 0.009	0.4258 ± 0.013	0.4504 ± 0.015	2.2015 ± 0.131	2.0740 ± 0.140
	ProtSCAPE	**0.2127 ± 0.005**	**0.1424 ± 0.006**	**0.1579 ± 0.002**	**0.9370 ± 0.125**	**0.9449 ± 0.125**
6JV8 (Chain A)	AlphaFlow	1.1663 ± 0.012	0.6576 ± 0.015	0.9631 ± 0.007	7.8291 ± 0.179	7.8324 ± 0.175
	MDGen	0.4729 ± 0.014	0.3818 ± 0.017	0.2787 ± 0.011	2.4517 ± 0.182	2.4535 ± 0.188
	ProtSCAPE	**0.1980 ± 0.003**	**0.1298 ± 0.004**	**0.1494 ± 0.003**	**0.4991 ± 0.067**	**0.5173 ± 0.064**
7LP1 (Chain A)	AlphaFlow	1.2303 ± 0.017	0.7533 ± 0.013	0.9727 ± 0.013	3.8058 ± 0.153	3.7298 ± 0.140
	MDGen	0.4694 ± 0.014	0.3623 ± 0.014	0.2983 ± 0.009	0.8995 ± 0.180	1.1783 ± 0.111
	ProtSCAPE	**0.1916 ± 0.010**	**0.0847 ± 0.009**	**0.1718 ± 0.007**	**0.3511 ± 0.091**	**0.3594 ± 0.091**
7JFL (Chain C)	AlphaFlow	1.9544 ± 0.050	1.5941 ± 0.028	1.1305 ± 0.048	13.0873 ± 0.352	13.0587 ± 0.351
	MDGen	0.3431 ± 0.003	0.2088 ± 0.007	0.2721 ± 0.003	**0.5640 ± 0.043**	**0.5820 ± 0.041**
	ProtSCAPE	**0.2117 ± 0.006**	**0.1691 ± 0.006**	**0.1273 ± 0.003**	0.7591 ± 0.070	0.7569 ± 0.071

5 Conclusions

In this work, we evaluated ProtSCAPE as a generative model for reconstructing protein conformational ensembles from sparsely sampled MD trajectories. While prior studies established its ability to generate structurally plausible conformations, our results show that ProtSCAPE also reproduces the underlying ensemble statistics of MD simulations with high fidelity.

Across five benchmark protein systems, ProtSCAPE consistently achieved strong structural quality, yielding the best MolProbity scores in every case while maintaining low clash scores and minimal Ramachandran outliers. More importantly, it outperformed competing approaches on most ensemble-level metrics, including RMSF correlation, RMWD, and PCA-based Wasserstein distances. These results indicate accurate recovery of residue-level flexibility, global conformational diversity, and collective motions that define the free-energy landscape.

Taken together, these findings suggest that ProtSCAPE offers a framework for MD trajectory upsampling and efficient exploration of protein dynamics. By combining geometric representation learning with an energy-regularized latent-space upon which to run diffusion, the model generates ensembles that are both physically realistic and distributionally consistent with reference simulations. More broadly, this supports the use of learned latent generative models as scalable tools for studying biomolecular dynamics beyond what is feasible with conventional simulation alone.

6 Future Work

While the present results demonstrate that ProtSCAPE performs strongly on a representative set of benchmark proteins, several important directions remain for future investigation. First, expanding evaluation to a broader and more diverse collection of test systems would provide a clearer picture of the model's generalization capabilities. The proteins considered here are relatively modest in size and structural complexity; future benchmarks should include larger multidomain proteins, membrane proteins, intrinsically disordered regions, and systems exhibiting substantial allosteric rearrangements. Such systems introduce more rugged free-energy landscapes and longer-range couplings, providing a more stringent test of whether ProtSCAPE can scale to biologically realistic conformational complexity. In addition to increasing system size, future studies should examine a larger number of proteins spanning distinct folds, functions, and dynamical regimes. Including systems with known experimentally characterized transitions could also help connect model outputs to biological function.

Another key extension is the incorporation of explicit side-chain modeling. The present study primarily evaluates backbone or Cα-level structural agreement, which captures large-scale conformational motions but omits many atomistic details critical for function. Side-chain orientations govern hydrogen bonding, salt bridges, catalytic geometry, ligand recognition, and packing interactions, and therefore represent an essential component of realistic ensemble generation. Extending ProtSCAPE to model full-atom structures or high-quality side-chain reconstruction would substantially broaden its practical utility.

Once side chains are included, a richer set of structural and biophysical analyses becomes possible. Future work could evaluate side-chain rotamer recovery, hydrogen-bond network preservation, solvent-accessible surface area distributions, residue-contact probabilities, and pocket-shapes relevant to ligand binding. Additional energetic analyses, such as approximate free-energy estimation or comparison of interaction networks across generated ensembles, would further test whether sampled conformations remain physically meaningful at atomistic resolution. Together, these directions would help establish ProtSCAPE not only as a tool for trajectory upsampling, but as a more general framework for efficient exploration of biomolecular free-energy landscapes.

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